



MILESTONE 1 PROGRESS PRESENTATION

S.H.I.E.L.D.

System for High Integrity Elimination of Low-Altitude
Drones

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Faculty Advisor: Dr. Madhur Tiwari · Client: Dr. Firat Irmak

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Milestone 1 Progress

Task	Status	Remaining
Compare and select technical tools	Done	None
"Hello world" demos	Done	None
Resolve technical challenges	90%	AI.Panther access for training
Compare and select collaboration tools	Done	None
Requirement Document	Done	None
Design Document	Done	None
Test Plan	Done	None

Technical Tools Selected

EDGE COMPUTER

NVIDIA Jetson Orin Nano Super

67 TOPS GPU. Reuses the unit on Dr. Tiwari's drone, replacing the unvalidated Hailo-8 plan.

CAMERA

e-CAM25_CUNOX (AR0234)

Global shutter avoids motion skew on fast targets; 1080p at up to 120 fps.

DETECTOR

YOLO26s

Chosen from a weighted matrix of YOLOv8, v11 and v26 (n/s/m). NMS-free and built for edge devices.

TRACKING

ByteTrack + Kalman filter

Persistent IDs through brief occlusion, with constant-velocity smoothing and prediction.

EMULATION

Unity 6 virtual camera

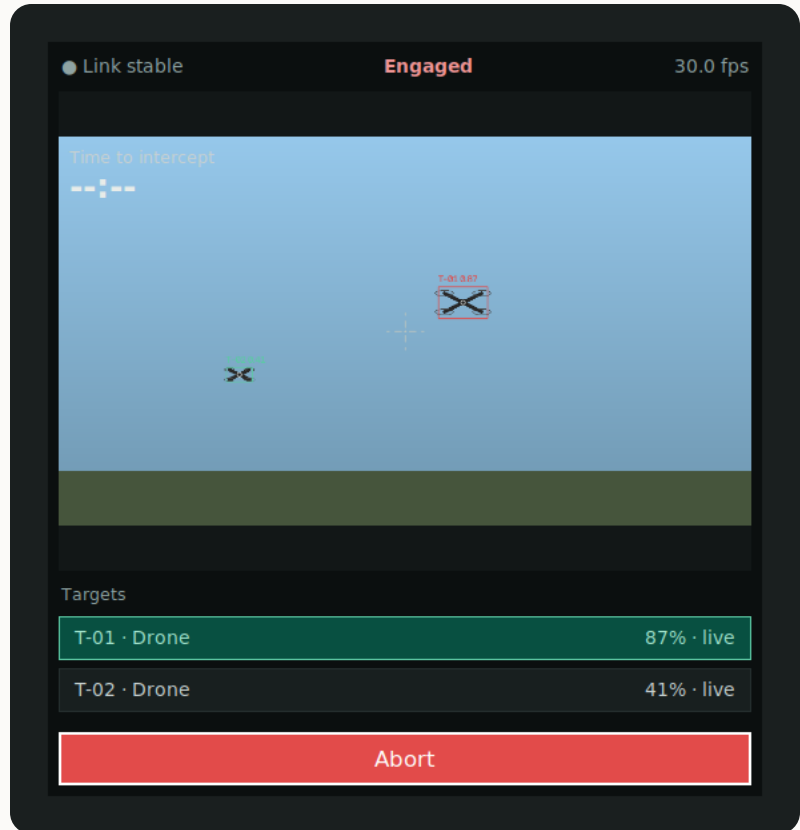
Streams rendered frames over TCP so the pipeline runs before the hardware exists.

TESTING

STE (WinUI3 + VB.NET)

Runs scripted, requirement-traced test cases and logs PASS/FAIL for every check.

"Hello World" Demos



- End-to-end emulation pipeline: YOLO → ByteTrack → Kalman filter, on a webcam, a video file, or the Unity virtual camera.
- Custom two-class model (drone, balloon) replaces the stock COCO weights; a yolo11n stand-in runs until YOLO26s is retrained.
- Ground control station window: live HUD, target list, and Engage/Abort gated at 25% confidence (AVS.03).

9,738

training images: three Kaggle sets plus 2,000 Unity-rendered synthetic frames

Collaboration Tools



GitHub

Version control and the project website.



Microsoft Teams

Communication with the AEE team.



Visual Studio

Python pipeline, C# STE app, VB.NET test scripts.



OneDrive + Docs

Shared team files; documents documents drafted and exported to PDF.

Object Detection Model Comparison

Criteria	Wt.	EfficientDet-Lite0		EfficientDet-Lite1		EfficientDet-Lite2		SSD MobileNet V1		SSDLite MobileDet		YOLOv8n		YOLOv8s		YOLOv8m	
		Rating	Wtd.	Rating	Wtd.	Rating	Wtd.	Rating	Wtd.	Rating	Wtd.	Rating	Wtd.	Rating	Wtd.	Rating	Wtd.
Accuracy (mAP50-95, INT8)	70%	1	0.70	2	1.40	2	1.40	0	0.00	1	0.70	2	1.40	4	2.80	4	2.80
Inference speed (ms/img, INT8)	30%	4	1.20	3	0.90	3	0.90	4	1.20	4	1.20	4	1.20	4	1.20	0	0.00
Total	100%		1.90		2.30		2.30		1.20		1.20		2.60		4.00		2.80

Rating scale: 0 Unsatisfactory · 1 Just Tolerable · 2 Adequate · 3 Good · 4 Very Good. Weighted = rating × weight.

RESULT

YOLOv8s scored highest (4.00), so the YOLO family was carried forward to a version and size comparison.

[1] <https://arxiv.org/html/2409.16808v1>

YOLO Model & Size Comparison

Criteria	Wt.	YOLOv8n		YOLOv8s		YOLOv8m		YOLO11n		YOLO11s		YOLO11m		YOLO26n		YOLO26s		YOLO26m	
		Rating	Wtd.	Rating	Wtd.	Rating	Wtd.	Rating	Wtd.	Rating	Wtd.	Rating	Wtd.	Rating	Wtd.	Rating	Wtd.	Rating	Wtd.
Accuracy (mAP50-95, INT8)	35%	0	0.00	1	0.35	2	0.70	0	0.00	2	0.70	2	0.70	2	0.70	3	1.05	4	1.40
Inference speed (ms/img, INT8)	15%	4	0.60	3	0.45	0	0.00	4	0.60	3	0.45	1	0.15	4	0.60	3	0.45	1	0.15
Headroom at 120 fps (peak)	15%	4	0.60	2	0.30	0	0.00	4	0.60	3	0.45	0	0.00	4	0.60	3	0.45	0	0.00
Headroom at 30 fps (min)	35%	4	1.40	3	1.05	0	0.00	4	1.40	3	1.05	1	0.35	4	1.40	4	1.40	3	1.05
Total	100%		2.60		2.15		0.70		2.60		2.65		1.20		3.30		3.35		1.90

Rating scale: 0 Unsatisfactory · 1 Just Tolerable · 2 Adequate · 3 Good · 4 Very Good. Weighted = rating × weight.

DECISION

YOLO26s scored highest (3.35) and was selected as SHIELD's detector.

[2] <https://docs.ultralytics.com/compare/yolov8-vs-yolo11>

[3] <https://docs.ultralytics.com/compare/yolov8-vs-yolo26>

[4] <https://docs.ultralytics.com/guides/nvidia-jetson>

[5] <https://www.ultralytics.com/blog/ultralytics-yolo26-vs-yolo11-vs-yolov8-which-one-should-you-use>

Edge Computer Comparison

Criteria	Weight	NVIDIA Jetson Nano		NVIDIA Jetson Orin Nano		Raspberry Pi 5 + AI HAT	
		Rating	Weighted	Rating	Weighted	Rating	Weighted
Compute Power	45%	1	0.45	4	1.80	3	1.35
RAM	25%	2	0.50	4	1.00	4	1.00
Power Draw	5%	4	0.20	2	0.10	3	0.15
Size/Mass	15%	3	0.45	3	0.45	4	0.60
Cost	10%	3	0.30	2	0.20	4	0.40
Total	100%		1.90		3.55		3.50

DECISION

Jetson Orin Nano selected for SHIELD to support real-time target detection and tracking.

Rating scale: 0 Unsatisfactory · 1 Just Tolerable · 2 Adequate · 3 Good · 4 Very Good. Weighted = rating × weight.

[6] <https://www.nvidia.com/en-us/autonomous-machines/embedded-systems/jetson-nano/product-development/>

[7] <https://www.nvidia.com/en-us/autonomous-machines/embedded-systems/jetson-orin/nano-super-developer-kit/>

[8] <https://www.raspberrypi.com/products/ai-hat/>

Camera Model Comparison

Criteria	Weight	Raspberry Pi Camera Module 3		e-CAM25_CUNOX		Arducam AR0234	
		Rating	Weighted	Rating	Weighted	Rating	Weighted
Interface/Compatibility	30%	2	0.60	4	1.20	3	0.90
Shutter Performance	25%	1	0.25	4	1.00	4	1.00
Resolution	15%	4	0.60	2	0.30	2	0.30
Frame Rate Capability	20%	3	0.60	4	0.80	3	0.60
Cost	10%	4	0.40	3	0.30	3	0.30
Total	100%		2.45		3.60		3.10

DECISION

e-CAM25 selected based on its Jetson compatibility, global shutter, and high frame-rate capabilities.

Rating scale: 0 Unsatisfactory · 1 Just Tolerable · 2 Adequate · 3 Good · 4 Very Good. Weighted = rating × weight.

[9] <https://www.raspberrypi.com/products/camera-module-3/>

[10] <https://www.e-consystems.com/nvidia-cameras/jetson-orin-nx-cameras/full-hd-ar0234-global-shutter-camera.asp>

[11] <https://docs.arducam.com/Nvidia-Jetson-Camera/Jetvariety-Camera/AR0234/>

Carrier Board Options

Key Criteria	Importance	Reference Carrier Board (datum)	Replacement Carrier Board	Integrated Carrier Board (Orin + FC)
Jetson Orin Nano Compatibility	5	datum	S	S
e-CAM25 Compatibility	5	datum	S	S
Flight Controller Integration	3	datum	S	+
Size	5	datum	+	–
Mass	5	datum	+	–
Cost	3	datum	–	–
Sum of Positives			2	1
Sum of Negatives			1	3
Sum of Sames			3	2
Weighted Sum of Positives			10	3
Weighted Sum of Negatives			3	13
Total			7	–10

Compared with the reference board: + better · S same · – worse. Weighted sums use the importance rating.

RESULT

Replacement Carrier Board considered to reduce size and mass while maintaining required compatibility.

Replacement Carrier Board Comparison

Criteria	Weight	Hiwonder Mini Carrier		Forecr DSBOARD-ORNXS		Seeed Studio J401	
		Rating	Weighted	Rating	Weighted	Rating	Weighted
Hardware Compatibility	20%	4	0.80	4	0.80	4	0.80
Flight Controller Integration	10%	4	0.40	4	0.40	4	0.40
Size	25%	2	0.50	4	1.00	3	0.75
Mass	25%	3	0.75	4	1.00	1	0.25
Cost	20%	4	0.80	2	0.40	4	0.80
Total	100%		3.25		3.60		3.00

RESULT

DSBOARD-ORNXS considered as a potential replacement carrier board to support payload integration.

Rating scale: 0 Unsatisfactory · 1 Just Tolerable · 2 Adequate · 3 Good · 4 Very Good. Weighted = rating × weight.

[12] <https://www.hiwonder.com/products/mini-carrier>

[13] <https://www.forecr.io/products/dsboard-ornxs>

[14] <https://www.seeedstudio.com/reComputer-J401-Carrier-Board-for-Jetson-Orin-NX-Orin-Nano-p-5636.html>

[15] https://developer.download.nvidia.com/assets/embedded/secure/jetson/orin_nano/docs/Jetson-Orin-Nano-DevKit-Carrier-Board-Specification_SP-11324-001_v1.3.pdf

Camera Mount/Gimbal

Criteria	Weight	Fixed Camera		1-Axis Gimbal		2-Axis Gimbal		3-Axis Gimbal	
		Rating	Wtd.	Rating	Wtd.	Rating	Wtd.	Rating	Wtd.
Cost	5%	4	0.20	3	0.15	2	0.10	1	0.05
Stabilization	35%	1	0.35	2	0.70	3	1.05	4	1.40
Tracking Complexity	35%	4	1.40	3	1.05	2	0.70	1	0.35
Mass	15%	4	0.60	3	0.45	1	0.15	0	0.00
Power Consumption	10%	4	0.40	3	0.30	2	0.20	1	0.10
Total	100%		2.95		2.65		2.20		1.90

DECISION

SHIELD will deploy with a fixed camera, as the improved video stability does not make up for the trade-offs in every other category.

Rating scale: 0 Unsatisfactory · 1 Just Tolerable · 2 Adequate · 3 Good · 4 Very Good. Weighted = rating × weight.

[16] <https://speedyfpv.com/products/2-axis-brushless-gimbal-for-fpv-camera-drones-lightweight-cnc-aluminum>

[17] <https://speedyfpv.com/products/3-axis-brushless-gimbal-for-fpv-camera-drones-lightweight-cnc-aluminum>

[18] <https://speedyfpv.com/products/pan-tilt-2-axis-servo-gimbal-for-fpv-quadcopter-car-plane-requires-2pcs-9g-servo>

[19] <https://www.waveshare.com/sg90-servo.htm>

[20] <https://www.basecamelectronics.com/simplebgc/>

Requirements: AVS Subsystem

ID	Requirement	Rationale
AVS.01	The drone shall be capable of detecting a target within 20 meters.	FPV drones carry 0.3–5 kg of explosives, and the DHS evacuation distance for a 5 lb explosive is ~ 21 m.
AVS.02	The drone shall recognize and track a target with a minimum span of 0.25 m.	The most commonly used kamikaze drones have a span of ~ 0.25 m or larger.
AVS.03	The drone shall maintain at least 25 percent confidence while en route to the target.	Provides continuous verification after the operator approves engagement, allowing an abort on a false positive if the track degrades.
AVS.04	The drone's onboard computing system shall process target detection within 33.33 ms of receiving the data from the onboard vision system.	Real-time detection is particularly important for an effective C-UAS. This will ensure that the system is acting on current information.
AVS.05	The drone shall receive engage and abort commands from the Ground Control System.	A "control station" is defined in 14 CFR Part 107, and there must be a remote pilot in command.
AVS.05.01	The drone shall send a heartbeat to the Ground Control System at a rate of 1 Hz.	The Ground Control System must have the status of the drone at all times. 1 Hz is the standard rate.

Functional: AVS.01–02 · Performance: AVS.03–04 · Interface: AVS.05, AVS.05.01. Each is traced to at least one STE test case.

Test Plan

Three stages

1 • Emulation

STE relaunches Unity and the pipeline for every case every case and logs each check against its requirement.

2 • Hardware bench (M2)

Same scripts and timing checks on the Jetson Orin Orin Nano.

3 • Flight

Balloon under a target drone in the Vertex cage: stationary, then vertical, then lateral.

Test case	Status
T01 Detection range & size	Automated
T02 False positives	Automated
T03 Confidence en route	Automated
T04 Detection latency	Automated (PC); Jetson in M2
T05 Track persistence	Planned, M2
T06 Engagement gating	Manual now; script in M2
T07 Abort, disengage, override	Stub; needs telemetry link
T08 Heartbeat & link loss	Link loss done; heartbeat stub
T09 Sources & platform modes	Planned, M2
T10 Moving target	Unity in M2; flight in spring

Technical Challenges

Pending

Training compute

My GPU is too slow to train YOLO26s model with large dataset. Access to the AI.Panther HPC cluster is requested, submitted by Dr. Irmak.

Planned, M2

Unproven edge hardware

Jetson Orin Nano benchmarks exist for YOLO, but I still have to test our entire detection pipeline on hardware to catch any issues early.

Planned, M2

Sustained real-time

Worst-case frame time and thermal throttling over a 6-minute flight, not just the average, must stay under 33.33 ms.

Feedback Incorporated

Client: Dr. Irmak

- Reuse the existing Jetson Orin Nano, no \$1,000 computer → **adopted**
- Camera of at least 30 fps → **AVS.04 budget**
- Manual takeover at any time → **GCS design**
- Two disengage modes: land or shut down → **GCS, M2**
- Balloon target: stationary, vertical, lateral → **test plan T10**

Advisor: Dr. Tiwari

- Develop detection and GNC separately, integrate at the end → **interface defined in M2**
- Research classical CV such as template matching → **reviewed; YOLO26s kept as detector**

Milestone 2 Plan (Oct 26)

01

Complete STE

Finish the test harness and write test scripts for every AVS requirement.

02

Train YOLO26s

Retrain the drone/balloon model on AI.Panther.

03

Benchmark on Jetson

TensorRT INT8; worst-case latency and thermal throttling.

04

Moving targets

Spawning and motion scripts in the Unity test scenes.

05

Manual override

GCS controls, with land and shut-down modes.

06

GNC interface

Define the target-state message sent to GNC GNC at 30 Hz.

Questions?

brysonnoble.github.io/SHIELD